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# Digital Transformation in Agriculture and Enhancing Food Security: Presenting Some Successful International Experiences

# التحول الرقمي في الزراعة وتعزيز الأمن الغذائي: عرض بعض التجارب الدولية الناجحة

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**Abstract:**

This study aims to highlight the importance of digital transformation in the agricultural sector and its role in enhancing food security, by presenting some successful international experiences from the Netherlands and China. The study explains how smart agriculture, including vertical farming, precision agriculture, the use of digital technologies, and agricultural data management systems, contributed to increasing crop productivity, reducing food loss, and improving the efficiency of natural resource utilization, thereby enhancing agricultural sustainability.

The results highighted that the success of digital transformation in the agriculutural sector in these countries is linked to several fundamental factors, most notably the adoption of supportive policies, the development of digital infrastructure, capacity building, and strengthening cooperation among various stakeholders.

**Keywords:** Digital Transformation; Smart Agriculture; Food Security; Agricultural Sustainability; International Experiences.

**JELClassificationCodes** : O57, O33, Q16, Q18.

**ملخص:**

تهدف هذه الدراسة إلى إبراز أهمية التحول الرقمي في القطاع الزراعي ودوره في تعزيز الأمن الغذائي، وذلك من خلال عرض بعض التجارب الدولية الناجحة لكل من هولندا والصين. وتوضح الدراسة كيف ساهمت الزراعة الذكية بما في ذلك الزراعة العمودية، الزراعة الدقيقة، واستخدام التقنيات الرقمية وأنظمة إدارة البيانات الزراعية، في رفع إنتاجية المحاصيل وتقليل الفاقد الغذائي، وتحسين كفاءة استغلال الموارد الطبيعية، بما يعزز الاستدامة الزراعية.

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وقد أبرزت النتائج أن نجاح التحول الرقمي في القطاع الزراعي في هذه الدول يرتبط بعوامل أساسية، أهمها تبني سياسات داعمة، وتطوير البنية التحتية الرقمية، وبناء القدرات البشرية، وتعزيز التعاون بين مختلف أصحاب المصلحة.

**كلمات مفتاحية**: التحول الرقمي، الزراعة الذكية، الأمن الغذائي، الاستدامة الزراعية، التجارب الدولية.

**تصنيفات JEL**: O57, O33, Q16, Q18.

## INTRODUCTION

Global food and agriculture systems face natural, social, economic, and political challenges, including limited land and water resources, climate change, rising production costs, supply chain disruptions, and demographic shifts, which require intensified production and more efficient, sustainable practices. In this context, innovation in the agricultural sector and digital transformation have become among the most prominent tools capable of creating a qualitative leap in agricultural production and its management. However, a comprehensive understanding of the impact of these technologies on achieving food security remains limited, especially in light of the technological gap between developed and developing countries. Hence emerges the importance of studying successful international experiences in this field, including the Netherlands' experience in high-tech agriculture and China's adoption of smart agriculture.

Analyzing these experiences provides a deeper understanding of the factors that determine the success of digital transformation in agriculture and enables the extraction of lessons and recommendations that can support national and regional efforts toward achieving sustainable food security.

### 1.1 Study Problem:

**How can countries, with their varying levels of economic development, employ digital transformation in agriculture effectively and sustainably to enhance food security?**

### 1.2 Hypotheses:

- Digital transformation in agriculture contributes to the optimal use of natural resources, thereby supporting agricultural sustainability and food security.
- The international case studies under review have achieved successes proving that adopting digital technologies in agriculture effectively contributes to improving productivity and resource utilization efficiency, thus enhancing food security.
- The success of applying digital transformation in agriculture depends on technological infrastructure and government support.

**1.3 Study Significance:** The significance of the study lies in highlighting the role of digital transformation in developing the agricultural sector and enhancing food security, by presenting successful international experiences with the aim of providing applicable insights and offering innovative solutions to support the sustainability of the agricultural sector. It also analyzes the challenges hindering agricultural digital transformation and proposes ways to enhance the efficiency and sustainability of agricultural production.

**2. Digital Transformation in Agriculture:**

Digital transformation involves leveraging digital technologies to create value and optimize benefits. It enhances production efficiency, reduces costs and greenhouse gas emissions, and ensures the optimal use of available resources.

Most economic sectors, including the agricultural sector, are witnessing radical transformations due to digital technologies. Digitization is considered a promising approach with the ability to bring change to this sector by making food production, both plant and animal, more efficient and environmentally friendly. It also contributes to enhancing resilience to challenges and improving the quality of life in rural areas (Florez, 2023, p8).

Since the 1990s, a wide range of digital technologies began to spread in the agricultural sector. However, the emergence of the concept of “smart agriculture” did not take shape until the second decade of the third millennium, when this concept became organized and institutionalized, presented as a promising technical-scientific solution to address the environmental, social, and economic crises facing the agricultural sector (Eleonore, 2025, p. 4).

**2.1 Definition of Smart Agriculture:** Smart agriculture, as presented by the Food and Agriculture Organization of the United Nations (FAO) during the Hague Conference in 2010 on Agriculture, Food Security, and Climate Change, refers to an integrated approach aimed at achieving food security and addressing the challenges of climate change simultaneously. This approach is based on integrating the economic, social, and environmental dimensions of sustainable development through three main pillars (FAO, 2025):

- Increasing agricultural productivity and sustainably raising farmers’ incomes;
- Enhancing the ability of agricultural systems and rural communities to adapt and be resilient to the impacts of climate change;
- Reducing greenhouse gas emissions or enhancing their ability to absorb them whenever possible.

Smart agriculture is not merely a set of ready-to-apply practices suitable everywhere, but rather an integrative approach that includes multiple elements designed and implemented according to local specificities. This approach includes measures within and outside farms, integrating technologies, policies, institutions, and investments to achieve the desired goals. The smart agricultural system consists of several main components, most notably(FAO, 2025):

- Management of farms, crops, livestock, aquaculture, and fisheries in a way that balances livelihood needs and food security in the short term, with climate change adaptation and mitigation priorities;
- Management of ecosystems and landscapes to ensure the preservation of essential environmental services that support food security, agricultural development, adaptation, and mitigation;
- Services directed to farmers and land managers enabling them to better manage climate risks and impacts, and to take effective measures to mitigate emissions;
- Changes at the level of the entire food system, including demand-oriented measures and interventions in value chains that enhance the benefits provided by climate-smart agriculture.

**2.2 The Importance of Smart Agriculture:** According to the Arab Organization for Agricultural Development, the importance of smart agriculture lies in its ability to enhance monitoring of field and climatic conditions, optimize the use of inputs and machinery, and boost crop productivity while preserving natural resources, particularly water. It also contributes to better cost management and project profitability, faster decision-making through digital tools, improved communication with farmers, stronger market linkages, and efficient livestock monitoring and disease control. (Kellou, 2024, p. 15)

**2.3 Technologies Used in Smart Agriculture:** In the past, agriculture relied on human and animal power, but technological development gradually changed agricultural production patterns, starting with the introduction of mechanization such as tractors and harvesters, and reaching advanced digital technologies that made agriculture smarter and more efficient. These technologies are divided into three main categories: automation and data collection technologies, data transmission technologies, and data processing technologies. The following table presents several technologies used in smart agriculture.

**Table 1.** Technologies Used in Smart Agriculture

| No. | Technical Category | Examples and Tools | Main Uses in Agriculture |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 01 | Navigation Technologies | GNSS systems such as GPS, GLONASS, Galileo, BeiDou | Location determination, automatic guidance, agricultural mapping |
| 02 | Agricultural Machinery Technologies | - GPS-guided driverless tractors- UAVs (Unmanned Aerial Vehicles)- UGVs (Unmanned Ground Vehicles) | Plowing and planting, pesticide spraying, aerial monitoring, data collection, harvesting, irrigation |
| 03 | Automated Mechanical Technologies | Control systems for engines, pumps, fans, and lighting | Controlling the environment of greenhouses and farms according to conditions |
| 04 | Mapping Technologies | Geographic Information Systems (GIS), remote sensing, satellite imagery | Soil maps, elevations, vegetation cover, crop productivity |
| 05 | Data Transmission Technologies | ZigBee, Bluetooth, Wi-Fi, GPRS/GSM, LoRa, Sigfox | Connecting sensors and networks, transmitting data from fields to processing systems |
| 06 | Data Processing and Artificial Intelligence | Genetic algorithms, deep learning, fuzzy logic, data mining | Agricultural data analysis, crop prediction, decision support |
| 07 | Wireless Sensor Networks (WSN) | Surface or underground networks (WUSN) | Monitoring soil moisture, local climate, supporting precision irrigation systems |
| 08 | Agricultural Internet of Things (IoT) | Sensors, robots, smartphone applications | Remote monitoring, weather forecasting, irrigation management, pest control, storage management |

**Source:** Prepared by the researchers based on: (Hassina, 2020, pp. 12-14)

These technologies represent the backbone of smart agriculture, as they provide accurate data that help farmers improve productivity, reduce waste, rationalize the use of resources, and enhance the ability to face environmental and climatic challenges.

## 3. Food Security:

**3.1 Definition of Food Security and Its Dimensions:** The definition of food security and its relationship to nutrition has evolved over the past five decades. In 1996, the World Food Summit proposed a definition of food security that is still in use today: *“Food security exists when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food to meet their dietary needs and food preferences for an active and healthy life”*(FAO, 2008, p. 01).

This definition is based on four fundamental dimensions internationally agreed upon, which must be addressed simultaneously in order to achieve food security objectives (FAO &WB, 2020, pp. 18–19):

- **Food availability:** linked to food supplies and the country's ability to provide sufficient nutritious food to meet population needs through local production or imports.
- **Food access:** the ability of individuals to obtain food physically and economically, meaning that people have the financial resources and means to acquire food.
- **Food utilization:** includes household- or individual-level preparation of food, its distribution within the household, and following a diverse and adequate diet—i.e., the effective use of food in terms of nutritional value, safety, and dietary quality.
- **Food stability:** the ability to access food and its continued availability over the long term, without being affected by shocks and crises such as climate fluctuations, market prices, or seasonality.

**3.2 Food Security in the World and Future Projections:** Global estimates indicate that hunger rates have decreased worldwide in recent years. In 2024, about 8.2% of the world's population suffered from hunger compared to 8.5% in 2023 and 8.7% in 2022. This means that around 673 million people were affected by hunger in 2024, representing a decrease of about 15 million compared to 2023 and 22 million compared to 2022. This improvement is attributed to noticeable progress in Southeast Asia, South Asia, and Latin America, while hunger persists in Africa. In 2024, 20.2% of people in Africa were affected by hunger, compared to 6.7% in Asia and 5.1% in Latin America and the Caribbean.

The prevalence of moderate or severe food insecurity worldwide has been declining since 2021, with about 28% of the global population in 2024 experiencing moderate or severe food insecurity, meaning they are unable to regularly obtain sufficient food (FAO et al, 2025, p. 11).

The 2025 SOFI report projects a decline in malnutrition by 2030, yet around 512 million people—60% in Africa—will still face hunger, leaving the world off track from achieving SDG 2 on ending hunger and promoting sustainable agriculture.

The decline has been exacerbated by the COVID-19 pandemic, sharp increases in food prices, and geopolitical disruptions. These crises pushed hunger and food insecurity rates to levels higher than before 2015, disproportionately burdening low-income groups and negatively impacting other development goals such as poverty reduction and health improvement.

Despite signs of recovery in recent years, persistent inflation has slowed progress and weakened purchasing power, limiting individuals' ability to access healthy diets (FAO et al, 2025, pp. 10–11).

**3.3 Factors Affecting Food Security:** Food security depends on multiple interconnected factors—natural, economic, social, and technological—that collectively determine people's ability to access adequate and safe food.

**3.3.1 Climate Change:** Climate change threatens agriculture by altering temperatures, rainfall, and increasing extreme events, pests, and diseases. As crop yields grow slower than population, food insecurity worsens. Projections show major declines in maize, rice, wheat, and other crops by 2030 and 2100, with global food prices expected to rise sharply.(Siva, Redmond, Sridhara&Rizan, 2023, pp. 4–5).

Thus, climate change is a fundamental threat to global agricultural systems, reducing crop yields and food security. Changes also result from human-induced factors such as pollution, urban expansion, industrialization, land-use change, and deforestation. These contribute to increased concentrations of water vapor, CO2, and other greenhouse gases in the atmosphere, accelerating climate change and leading to phenomena such as global warming. This calls for effective adaptation strategies and comprehensive mitigation efforts to reduce its negative impacts (Wijerathna& Pathirana, 2022, p. 2).

**3.3.2 Population Growth:** According to United Nations projections (2022), the world's population will reach 8.5 billion by 2030 and about 9.7 billion by 2050, requiring a 70% increase in food production, as reported by the UN Department of Economic and Social Affairs.

Global population growth is expected to continue over the next 50–60 years, reaching about 10.3 billion by the mid-2080s, up from 8.2 billion in 2024. This makes it essential for food systems to evolve to keep up with growing demand. Food security remains the cornerstone for ensuring a healthy future for all, through the provision of healthy and balanced food, adoption of sustainable farming practices, and ensuring fair distribution of food among all segments of society (United Nations, 2024).

**3.3.3 Scarcity and Degradation of Land and Water:** Global food systems are under growing pressure from land and water degradation and resource scarcity, threatening the ability to feed over 9 billion people by 2050. Although food production tripled in the past 50 years, much of it relied on unsustainable practices that harmed vital resources, now causing declining productivity in many regions worldwide.(Nacef&Keddaoui, 2024, p. 315).

**3.3.4 Energy Security and Food Prices:** Energy security is essential for the food system, as agriculture depends heavily on gas, electricity, and fuel, and producing inputs like fertilizers and pesticides requires substantial energy. Recent international crises have driven up energy prices, increasing production costs, disrupting supply chains, and raising global food prices. (Cherief, 2023, p. 16).

Food prices strongly affect global food security, especially in developing countries where household spending on food represents a high share of income. Research indicates that these fluctuations are due to multiple factors, including oil and energy price changes, increased demand for biofuels, climate factors, and trade policy impacts (Daniel, 2020, pp. 43–44).

According to a study conducted by FAO, UNCTAD, and several other international organizations, agricultural prices are highly sensitive to fluctuations, which calls for developing mechanisms for risk management and sharing between markets. The study also stresses that strengthening such mechanisms can provide more effective protection from sudden price shocks, particularly in developing countries with high proportions of poor rural households relying on self-produced food as a main survival source (FAO et al., 2011).

## 4. Presentation of Some Successful International Experiences in the Field of Digital Transformation in Agriculture

**4.1 The Netherlands Experience:** The Netherlands, despite its small size, is the world's second-largest exporter of agricultural products after the USA. Limited farmland and climatic challenges, such as rising water levels and flood risks, have driven the country to adopt smart agricultural technologies to maximize production on smaller areas. With high population density demanding sustainable food production near consumers, the Netherlands has pioneered initiatives like smart greenhouses, precision sensing, robotics, IoT, drones, big data, and AI.

**4.1.1 Vertical Farming in Urban Areas in the Netherlands:** Vertical farming uses a multi-layered approach to crop cultivation, allowing more efficient food production in terms of land use. The Netherlands, despite significant territorial constraints, leads in the development and implementation of sustainable vertical farming systems. This innovation helps improve the use of limited space and reduces the carbon footprint of traditional farming practices. Hydroponics is a core component of vertical farming and is especially important in urban areas in the Netherlands, where resource efficiency is a top priority. Hydroponic systems allow crops to grow in nutrient solutions without soil, reducing water consumption compared to traditional methods (Nina, Lucas&Sridar, 2024, pp. 106–107).

The adoption of vertical farming in the Netherlands responds well to climate change and other environmental challenges that often hinder traditional agriculture. Vertical farming can be carried out indoors, making it unaffected by weather conditions and ensuring year-round food production.

In a 2024 study by Nening Nina, Laura Lucas & Katja Sridar, which combined quantitative and qualitative elements for a comprehensive analysis of the effectiveness and sustainability of vertical farming in Dutch urban areas, covering a variety of agricultural operations applying modern hydroponic techniques, the results showed high resource efficiency, up to 85% water savings compared to traditional farming, and a significant increase in productivity per square meter. However, some crops do not adapt well to vertical systems, and energy consumption—particularly electricity—remains a major challenge, along with high operating and maintenance costs (Nina, Lucas &Sridar, 2024, pp. 106–107).

**4.1.2 Precision Agriculture in the Netherlands:** Precision agriculture uses technologies like GPS, sensors, and robotics to support farm-level decision-making, enabling more sustainable crop and livestock production with actions tailored to specific plots or even individual animals.

In the Netherlands, smart agriculture systems are being developed particularly in livestock farming, especially smart dairy farms. Wageningen University participates in several research projects, such as virtual fencing for borderless grazing, cows wearing sensors to monitor behavior, sensors measuring energy levels in farmed fish, and new robotic projects to improve livestock systems (Wageningen University, 2025). The Dutch government seeks to encourage farmers to adopt precision agriculture. In 2018, the National Precision Farming Initiative was launched to help farmers apply modern technologies (OECD, 2023, p. 133).

The Dutch government supports the broader digital transformation of the agricultural sector, not just precision farming, through an integrated set of tools and policies. It provides direct support to farmers via training vouchers (SABE) for digital skills acquisition, and supports related investments in farms through programs such as MIA and VAMIL. Experimental fields and model farms also contribute to testing agricultural innovations and preparing them for broader adoption. At the European level, the Netherlands supports the *Agriculture of Data (AgData)* partnership within Horizon Europe, employing remote sensing and AI technologies to improve agricultural sustainability and productivity, and enabling member states to monitor and evaluate their policies more accurately. The key criterion for adopting any new technology is the added value it provides in tackling climate change, protecting biodiversity, and supporting long-term agricultural sustainability (OECD, 2023, p. 133).

**4.1.3 WUNDER Project: Climate-Adaptive Water Production and Management Systems:**

The WUNDER project tackles the impacts of drought on Dutch agriculture and ecosystems, which are increasingly threatened by climate change. It develops integrated models to study soil–plant interactions during drought and designs adaptive strategies for water and land management. By working with farmers, water managers, and policymakers, WUNDER creates practical tools for drought monitoring and forecasting, helping decision-making at local and national levels. (NOW, 2022).

The University of Twente leads a consortium of 17 partners from the public and private sectors to develop an integrated monitoring and modeling system to understand soil and vegetation behavior during drought. The consortium received a €1.5 million grant from the Netherlands Organization for Scientific Research (NWO) to create innovative solutions to mitigate drought damage. WUNDER works on developing a new system using several smart technologies, including remote sensing, digital twins, integrated modeling, AI, data analysis, and IoT, to determine the best measures to reduce drought damage to crops (University of Twente, 2022).

**4.1.4 Food Security in the Netherlands:** The Netherlands' agricultural and food sector is highly productive and innovative, making it one of the world's most efficient and the second-largest agricultural exporter. Its strong trade position ensures national food security through production surpluses. However, challenges such as environmental sustainability, market volatility, nutrient surpluses, nitrogen leakage, and ammonia emissions threaten natural areas, water quality, and biodiversity.(OECD, 2023).

The OECD report of June 26, 2023, recommends adopting clear policies to address water and biodiversity degradation and nutrient surpluses by establishing clear environmental limits, building a data-driven smart agricultural sector, enhancing digital innovation, focusing on sustainable environmental objectives, and ensuring decisive action guided by a promising governmental long-term vision (OECD, 2023).

The Dutch success in smart agriculture stems from advanced technology, strong institutional support, innovation, and research, especially from Wageningen University. Through vertical farming, precision agriculture, and projects like WUNDER, the Netherlands has become a global model for sustainable agriculture, though continued efforts are needed to reduce environmental pressures and secure long-term food and resource sustainability.

**4.2 China's Experience:** By 2022, China's digital villages witnessed significant progress in digital infrastructure, with fifth-generation (5G) network coverage including all urban centers at the county level and the internet becoming available in various villages. The rural digital economy continued to grow, with the value of online sales reaching 2.17 trillion yuan. As for smart agriculture, the informatization rate in agricultural production reached 25.4%, and digital governance improved, with more than two-thirds of agricultural government services completed online, while the proportion of villages enjoying digital services exceeded 86% (Ministry of Agriculture and Rural Affairs, 2024).

At the level of practical applications of climate-smart agriculture, the agricultural water conservation project in Huaiyuan County, Anhui Province, showed tangible success, with more than two-thirds of farmland irrigated efficiently and water use efficiency increased by more than 50%, which contributed to enhancing the capacity to withstand droughts and floods (FAO, 2023).

On March 1, 2023, the Central Cyberspace Administration of China and the Ministry of Agriculture and Rural Affairs – Department of Market and Digital Transformation issued a report on the development of digital villages in China, with contributions from China Agricultural University, the China Academy of Information and Communications Technology, and Renmin University. The report reviews the progress made since the State Council launched the Digital Villages Development Strategy in 2019, noting the achievements of 2022 in rural industry, talent development, culture, environment, and institutional governance. The most important achievements include (DCZ-China, 2023):

**4.2.1 Digital infrastructure for villages:** The number of internet users in rural areas reached 293 million, and rural internet coverage reached 58.8% by June 2022, with 1.9 million 5G base stations covering 96% of counties and towns.

**4.2.2 Progress in smart agriculture:**

- The overall digitization rate of agriculture reached 25.4% in 2022, including production, harvesting, processing, marketing, and data management.

- The first national seed identification information system was established to track and share information on seeds.
- Digitization in large-scale agriculture reached 21.8% in 2021, with the highest rates for wheat (39.6%), rice (37.7%), and cotton (36.3%).
- Digital agriculture pilot fields and digital machinery extended to 150,000 mu (10,000 hectares) in 2022, with productivity increasing by 14.3%/mu and revenues rising by 500 yuan/mu, while the use of nitrogen fertilizers decreased by 32.5%, phosphorus by 16.8%, and other chemicals by 38%.
- The digitization rate in animal husbandry reached 34% in 2021, and 36.4% for the poultry sector, which improved labor productivity by 30%; digitization in fisheries and aquaculture reached 16.6% in 2021.
- State farms used 8,300 machines equipped with the BeiDou navigation system to cover 60 million mu (4 million hectares), with the digital revenues of the Beidahuang Group increasing by 8.7 billion yuan in 2021.
- The number of unmanned agricultural machines reached 121,000, operating over an area of 1 billion mu (66.7 million hectares).

**4.2.3 Rural digital economy:**

- The size of rural e-commerce reached 2.17 trillion yuan in 2022, an increase of 3.6% over the previous year.
- Digital technology contributed to the development of rural tourism and the provision of new income opportunities, with 11.2 million people returning to villages to create new projects in 2021, an increase of 10.9% over the previous year.
- The number of rural digital payment users reached 227 million by June 2022, with 17.3 billion and 576 billion digital banking and non-banking transactions, respectively, in 2021, up by 22.5% and 23.5%, respectively.

**4.2.4 Rural governance and social services:**

- 355 county-level platforms were established to provide digital government services and social governance, with enhanced digital emergency services covering 79.7% of rural areas.
- Digital education services included all compulsory primary and secondary schools, with 34,000 core educational materials and 22,000 materials for vocational education distributed.
- Digital health services reached 85% of municipalities and 69% of counties, while users of digital social security cards reached 619 million people.

**4.2.5 Digital environmental monitoring:** 3,005 air quality monitoring stations and 4,688 surface water quality monitoring stations were established in villages, fully integrated into the digital system.

**4.2.6 Government support:** legislative documents and official plans include the Rural Revitalization Law, the Fourteenth Five-Year Plan (2021–2025) for economic and social development, the National Digitization and Modern Agriculture Plan, and the Action Plan for Digital Villages Development (2022–2025).

Despite these achievements, the report points to gaps in digital infrastructure between the advanced coastal regions and the central and western regions, in addition to challenges such as low digital literacy among the rural elderly, and the need for further improvement of technology standards and data protection, as well as increased community participation to ensure sustainable and inclusive development (DCZ-China, 2023).

On October 25, 2024, China launched a five-year action plan for smart agriculture for the period 2024–2028, aimed at accelerating the digital transformation of the entire agricultural industry chain as part of measures to increase domestic food production. According to the Ministry of Agriculture of China, this plan will seek to establish a system for digital agricultural technologies, as well as a national platform for large agricultural and rural data by 2028, in order to enhance agricultural efficiency, increase productivity, and support national food security (Mei Mei Chu, 2024).

**4.2.7 Food security in China:** Food security in China is a government priority, with major efforts made to ensure stable food supplies and meet the needs of its 1.4 billion population. Grain production in China reached 695.41 million tons in 2023, an increase of 1.3% compared to the previous year, a record for the ninth consecutive year in which production exceeded 650 million tons. Wheat production reached 136.59 million tons, down 0.8%. To boost grain production, the central government raised the minimum purchase prices for wheat and rice in 2023, and improved support policies for farmers producing corn and soybeans.

According to the 2024 government report, China is committed to strengthening food security by improving grain production, storage, and processing, adopting modern technologies, and achieving self-sufficiency in staple grains. Nevertheless, challenges such as climate change and import dependence require ongoing innovation and international cooperation to maintain sustainable food security. (Ministry of Agriculture and Rural Affairs, 2024).

## 5. CONCLUSION

The study highlights digital transformation in agriculture as a strategic path to strengthen food security amid growing challenges. International experiences from the Netherlands and China show that digital innovations enhance efficiency, reduce waste, and support sustainability. Despite proven benefits, countries face barriers such as climate change, limited resources, weak infrastructure, and lack of supportive policies. Identifying success factors from global practices can guide the development of tailored strategies for different national contexts.

### 5.1 Study Results:

- Integrating key modern technologies and innovations such as artificial intelligence, vertical and precision agriculture, and big data analytics into the agricultural sector will significantly enhance production, ensure optimal resource utilization, and achieve sustainable food security.
- The study concluded that despite the differences in agricultural environments between the Netherlands and China, both countries achieved success, proving that adopting digital technologies in agriculture effectively contributes to improving productivity and resource-use efficiency, which in turn raised food security levels.
- The success of applying digital transformation is linked to several fundamental factors: government support, technological innovation, availability of advanced digital infrastructure, capacity building and training of farmers, and strengthening cooperation between the public and private sectors and stakeholders.

### 5.2 Suggestions:

- Balanced investment in digital infrastructure (internet, cloud computing, databases) and agricultural infrastructure (irrigation, storage, transport).
- The necessity of enhancing awareness of sustainable investment and supporting technological innovation in the agricultural sector, integrating artificial intelligence and big data technologies into agricultural systems to improve production forecasting and address climate risks.
- Developing continuous training programs targeting farmers and agricultural engineers to strengthen their digital skills and ensure optimal use of technology, while also focusing on protecting farmers from risks through reliance on smart insurance based on modern technologies.
- Encouraging partnerships between the public and private sectors to

support innovation and provide the necessary financing for digital agricultural projects.

- Supporting research and development to monitor best international practices and adapt them to the specificities of each country, benefiting from global expertise, and establishing regional cooperation in smart agriculture.

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